PRACTICAL - 7

AIM: Write a program to implement a queue using arrays and perform Enqueue, Dequeue and Display operations.

Program:

```
#include <stdio.h>
#include <stdlib.h>
#define SIZE 5

int queue[SIZE];
int front = -1, rear = -1;

void enqueue(int value) {
   if (rear == SIZE - 1) {
      printf("Queue Overflow\n");
      return;
   }

   else if(front == -1 && rear == -1){
      rear = 0;
      front =0;
      queue[rear] = value;
   }
   else{
      rear++;
       queue[rear] = value;
      printf("%d inserted to queue.\n", value);
   }
}

void dequeue() {
   int y;
   if (front == -1) {
```

```
        printf("Queue Underflow\n");
        return;
    }
    else{
     y = queue[front];
    }

    if (front == rear) {
        front = rear = -1;
    }
    else{
        front++;
    }
}

void display() {
    if (front == -1) {
        printf("Queue is empty.\n");
        return;
    }
    printf("Queue elements: ");
    for (int i = front; i <= rear; i++) {
        printf("%d ", queue[i]);
    }
    printf("\n");
}

int main() {
    int choice, value;

    while (1) {
```

```
        printf("\nQueue Operations Menu:\n");
        printf("1. Enqueue\n");
        printf("2. Dequeue\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
        case 1:
            printf("Enter value to enqueue: ");
            scanf("%d", &value);
            enqueue(value);
            break;

        case 2:
            dequeue();
            break;

        case 3:
            display();
            break;

        case 4:
            printf("Exiting program.\n");
            exit(0);

        default:
            printf("Invalid choice\n");
        }
    }
    return 0;
```

}

OUTPUT:

```
Queue Operations Menu:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter value to enqueue: 1

Queue Operations Menu:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter value to enqueue: 1
1 inserted to queue.

Queue Operations Menu:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 3
Queue elements: 1 1
```

```
Queue Operations Menu:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 2

Queue Operations Menu:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 3
Queue elements: 1

Queue Operations Menu:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 4
Exiting program.
```